



Seungyeop Lee

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RESEARCH INTERESTS

Wearable Robot

Soft wearable robot

Exoskeleton robot

Multimodal sensing

User-specific adaptation algorithms

EDUCATION

Korea University <i>M. S. in Mechanical Engineering. Integrated Major in Intelligent Mobility. GPA: 4.45/4.5</i> <i>Research Lab: Human Machine Systems Lab (Prof. Shinsuk Park)</i> <i>Thesis: A Development of Intuitive Controller and Monocular Vision based Interface for Disaster Response UAV</i>	Sep. 2022 - Feb. 2025 Seoul, Republic of Korea
Korea University <i>B. S. in Mechanical Engineering. GPA: 3.45/4.5</i> <i>(Note: Absence during Oct. 2017 – Jun. 2019 due to military service)</i>	Mar. 2016 - Aug. 2022 Seoul, Republic of Korea

RESEARCH EXPERIENCES

Robot Vacuum Cleaner Spatial AI Development <i>Full-time, Samsung Electronics, Suwon, South Korea</i> - Developed a computationally efficient vision-based scene graph generation pipeline for robotic vacuum cleaners, representing indoor rooms, objects, and their spatial relationships on resource-constrained platforms - Developed a Bayesian inference-based algorithm for room-type classification by aggregating semantic observations collected during robot navigation - Designed and optimized keyframe selection methods for efficient collection and processing of robot image data - Integrated VLM, perception, and navigation modules into a unified Spatial AI system and validated the integrated system using NVIDIA Isaac Sim simulation	April. 2026 – Present
Robot Vacuum Cleaner Mop Module Mechanism Design <i>Full-time, Samsung Electronics, Suwon, South Korea</i> - Prototyping design for detecting mop attachment/detachment in a robotic vacuum cleaner, with improved detection performance using a shielding plate - Structural design for enhanced module durability and development of performance evaluation methods - Contributed to the launch of the VR90F robotic vacuum cleaner, supporting development activities from design completion to mass production readiness - Managed mass production processes to ensure stable product quality, manufacturability - Developed and implemented a vision inspection system for production lines to control part dimensions and improve quality consistency	Feb. 2025 – April. 2026
Development of an Embedded Computer Controller based Soft Exo-suit <i>Intern, Wearable Robotics Laboratory(Advisor: Prof. Jinsoo Kim), Seoul National University, South Korea</i> - Developed a wearable robot controller based on the Nvidia Jetson Orin Nano platform - Developed motor and sensor data transmission algorithm under 10ms via CAN communication using multi-threading data processing - Design and Validation of a Sensor Integration Module	Dec. 2024 – Feb. 2025
Development of a Drone Control Assist Interface System in Non-GPS Environment <i>Master's Student, Human-Machine Systems Lab(Advisor: Prof. Shinsuk Park), Korea University, South Korea</i> - Developing a drone control assistance HMI system to prevent crashes due to human error during indoor drone flights - Utilized Robot Operating System (ROS) on Jetson Nano (companion computer) to send drone flight commands to Pixhawk (flight controller), enabling data transmission between the controller and the companion computer - Employed MAVLink system for communication between the computer and flight controller and used MAVROS package in ROS for command transmission - Testing flight assist algorithms using ROS and Gazebo simulation	Sep. 2022 – Dec. 2024
Monocular Depth Estimation Training Method using Synthetic Dataset <i>Intern Robotics Researcher, Intelligent Machine Perception and Learning Lab(Advisor: Prof. David Han), Drexel University, USA</i> - Used CycleGAN algorithm to convert data collected from simulations into images resembling real-world environments for training a depth estimation deep learning model - Suggest evaluation method comparing between predicted depth and ground truth(lidar point cloud) - Confirmed that the depth estimation performance of the supervised monocular camera depth estimation deep learning model trained with mixture of actual and converted synthetic datasets was superior to other results	Nov. 2023 - Feb. 2024
Development of Autonomous Lighter Than Air Vehicle <i>Robotics Researcher, Intelligent Machine Perception and Learning Lab(Advisor: Prof. David Han), Drexel University, USA</i> - Developed a drone attitude control system using STM32 embedded system and PID control - Detected destinations and targets using OpenCV library and YoloV5 - Created a suction device for capturing objects in the air	Sep. 2023 - Nov. 2023
Development of Tethered Drone System <i>Undergraduate Research Student, Human-Machine Systems Lab(Advisor: Prof. Shinsuk Park), Korea University, South Korea</i> Developed a tethered drone for integration with mobile robots in smart farms	Mar. 2021 - Aug. 2022

- Integrated using EtherCAT wired system, with a Windows PC as the main controller and Arduino as the motor driver

Coupling Structure for Autonomous Driving Vehicle and Simulation Environment for Driving Test

Mar. 2021 - Dec. 2021

Intern, Start-up, USA(Worked in South Korea)

- Built simulation roads using Unity and integrated Unity models into the LGSVL vehicle simulation platform
- Designed king pin-based coupling structure for vehicles and developed side leg structure to enhance vehicle stability

PAPERS AND CONFERENCES

Seungyeop Lee, Knut Petorson, Solmaz Arezoomandan, and David Han. "Domain-Transferred Synthetic Data Generation for Improving Monocular Depth Estimation", *2025 25th International Conference on Control, Automation and Systems (ICCAS)*. IEEE, 2025

Seungyeop Lee, and Shinsuk Park. "Obstacle Avoidance for Small Unmanned Aerial Vehicles Using Deep Learning-Based Monocular Depth Estimation", *Institute of Control, Robotics and Systems Conference (ICROS)*, Daejeon, Korea 07. 2024

Seungyeop Lee, and Shinsuk Park. "Development of a ROS-Based Indoor Autonomous Quadcopter Platform Using Visual Odometry", *Institute of Control, Robotics and Systems Conference (ICROS)*, Samcheok, Korea, 06. 2023

PROJECTS

Defend the Republic (Autonomous Drone Competition), *Intern Robotics Researcher*

Sep. 2024 - Nov. 2024

Lehigh University, USA

- Participated in an autonomous flying vehicle development competition, aiming to collect objects in the air and transport them to a destination
- Developed a lightweight flying vehicle utilizing helium gas buoyancy, responsible for creating the main body and the object-grabbing device
- Used carbon pipes and 3D-printed parts for construction; crafted the drone's main body from balsa wood and designed the object-grabbing device using small BLDC motors, carbon rods, and vinyl film

Korea Robot Aircraft Competition, *Researcher*

Apr. 2021 - Sep. 2021

Korea Aerospace Industries Association, South Korea

- Developed a pedestrian tracking drone using Pixhawk(Flight Controller) and Jeton Xavier(Companion Computer)
- Mounted a three-axis gimbal under the drone to compensate for inaccurate yaw control of drone and increased pedestrian detection rates
- Employed YOLOv5 for pedestrian detection and improved detection rates by training on a custom dataset
- Maintained and repaired the drone hardware system

Club Revival Project, *Club president*

Apr. 2021 - Jun. 2021

Korea University, South Korea

- Suggested ideas to revitalize non-face-to-face school clubs during COVID-19 period and promoted work-sharing efficiency by using time scheduling method

2020 Social Economy Engineering Ideas Competition, *Researcher*

Sep. 2020 - Oct. 2020

University of Seoul, South Korea

- Aimed to support reduced physical abilities in subway stations frequently used by the elderly
- Suggested ideas to address physical challenges faced by the elderly when using the subway, implemented in Fusion 360 3D CAD

Creative Comprehensive Design Competition, *Researcher*

Jun. 2020 - Sep. 2020

Innovation center for engineering education, Korea University, South Korea

- Developed an automatic separation and disposal trash bin to address the beverage cup recycling problem on campus
- Designed the exterior of the trash cans using acrylic plates and used DC motors to enable the separation of cups and their liquid contents

Pi village Start-up Team - DRONEDU, *Founder*

Jan. 2020 - Oct. 2020

Korea University, South Korea

- Developed a drone education program for high school and college students, while developing the idea at a start-up activity space within the campus

21st National University Student Model Aircraft Competition, *Team leader*

Nov. 2019 - Nov. 2020

Gyeongsang National University, South Korea

- Created a scaled model of a Boeing aircraft using materials like balsa wood, carbon fiber, carbon pipes, 3D printer, and foam board
- Developed a 3D CAD model of the aircraft's internal structure and built it using 3D printers and laser cutters
- Designed and fabricated a mount model for the DC motor's duct-fan shape using CAD software

Horang-Nabi(UAV club) foundation, *Founder*

Aug. 2019 - Dec. 2021

Korea University, South Korea

- Founded Korea University's First Model Aircraft Club
- Organized a club activities program focused on building Arduino-based drones and modifying the structure of model aircraft using DC motors
- Held Arduino drone and C++ group study session

AWARDS AND HONORS

Poster Competition of Results Report for Future Automobile Industry-university assignment Project <i>Excellence Award, Inter-university Alliance for Future Vehicular Technology (I4FT), South Korea</i> - Performed an industry-academic project to develop proprietary technology in Intelligent Mobility integrated major course - Developed a tethered drone using EtherCAT data transformation method and suggested combinable structure with other machines such as manipulator and mobile robot	Jan. 2023
21st National University Student Model Aircraft Competition <i>Third prize, Gyeongsang National University, South Korea</i> Developed a scale-down model Boeing aircraft with special interior structure with carbon pipe and 3D printer	Nov. 2020
2020 Social Economy Engineering Idea Competition <i>Third prize, University of Seoul, South Korea</i> Designed a foldable stair climbing aid machine to address social problems for the elderly when using the subway	Oct. 2020
2020 Creative Comprehensive Design Competition <i>Third prize, Innovation center for engineering education, Korea University, South Korea</i> Developed automatic recycling trash can to solve the recycling issue at school	Sep. 2020
22th Pi village Start-up Team - DRONEDU <i>First prize, Korea University, South Korea</i> Developed drone education contents for students	Jun. 2020

SKILLS AND TECHNIQUES

Hardware 3D CAD(NX, Fusion 360), Rapid prototyping(laser cutting, 3D printing)
Programming Python, C++, ROS, MATLAB, Microprocessor(Jetson, Arduino, Raspberry PI)

EXTRA EXPERIENCES

Teaching Assistant, Dynamics Explain Adams system simulation tools about how to build two link manipulators	Sep. 2024 - Nov. 2024
Teaching Assistant, Robotics Explain Adams system simulation tools, explain tutorial of Raspberry PI and show how to control DC motor with PID control	Mar. 2024 - Jun. 2024 Mar. 2023 - Jun. 2023

REFERENCES

- Dr. Shinsuk Park**
- Associate professor, Dept. of Mechanical Engineering, Korea University
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- Dr. David Han**
- Associate professor, Dept. of Electrical and Computer Engineering, Drexel University
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- Dr. Jinsoo Kim**
- Assistant professor, Dept. of Electrical and Computer Engineering, Seoul National University
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